

Exploring the universality of scaled particle spectra with Bayesian inference in heavy-ion collisions

Based on: [arXiv:2508.21176](https://arxiv.org/abs/2508.21176), work in progress with

(The ExTrEMe Collaboration)

Experiment and Theory in Extreme Matter

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RETINHA XXXII 2025 — October 23, 2025



HIGH-ENERGY NUCLEAR COLLISIONS

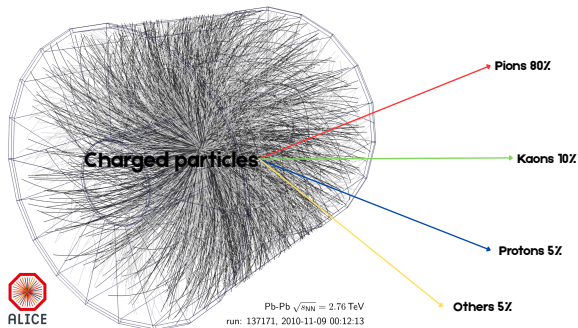
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- Investigate the fundamental properties of nuclear matter under extreme conditions;
- **Relativistic heavy-ion collisions:** these collisions yield a large number of particles $\sim$ up to 35,000/collision (LHC);
- Properties and correlations between these particles $\Rightarrow$ information about the dynamics of the generated fluid-like state (QGP).

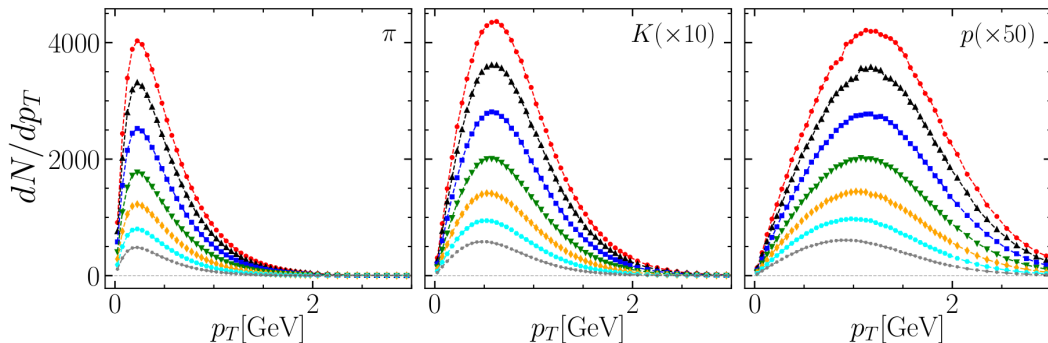


TRANSVERSE MOMENTUM DISTRIBUTION OF IDENTIFIED PARTICLES

Particle distribution in the transverse plane is one of the main observables to extract relevant information about QGP

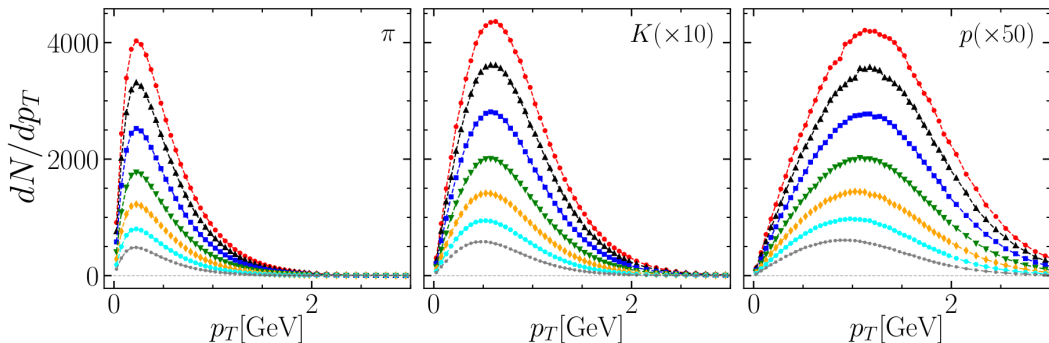
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$p_T = \sqrt{p_x^2 + p_y^2} \Rightarrow dN/dp_T$: count the number of particles (average over events) in a given p_T bin.

NEW OBSERVABLE: SCALING THE p_T -SPECTRA

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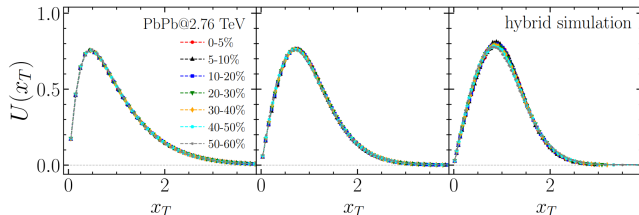
$$U(x_T) \equiv \frac{\langle p_T \rangle}{N} \frac{dN}{dp_T} = \frac{1}{N} \frac{dN}{dx_T}, \quad x_T = \frac{p_T}{\langle p_T \rangle}, \quad (1)$$

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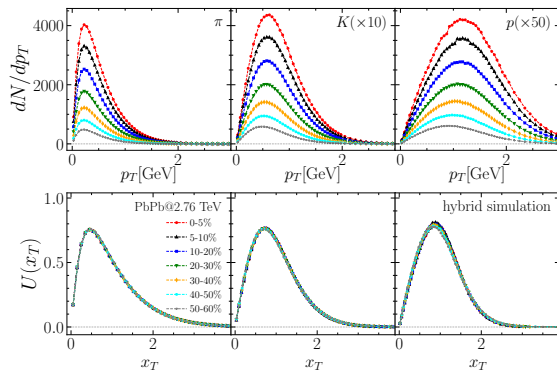
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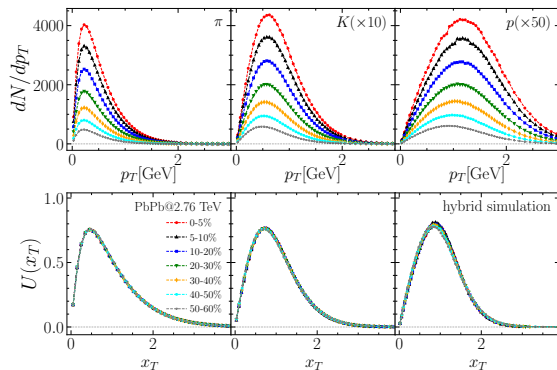


UNIVERSAL SCALING: UNIVERSALITY OVER CENTRALITY



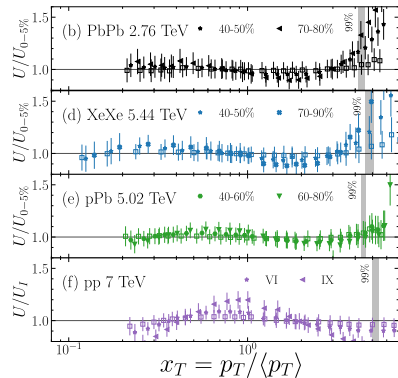
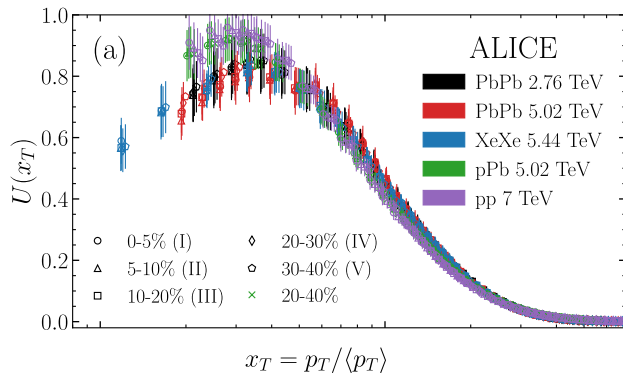
- All events approximately collapse onto the same curve! $\Rightarrow$ **Universality over centrality!**

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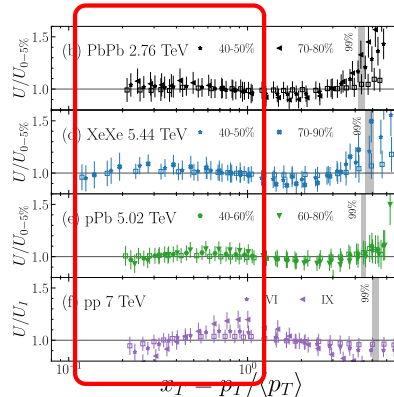
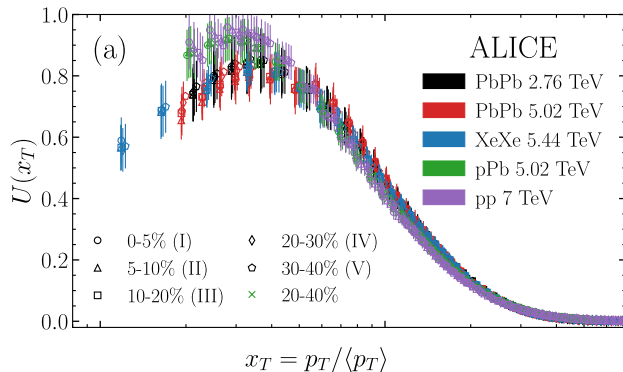


- All events approximately collapse onto the same curve! $\Rightarrow$ **Universality over centrality!**
- This is a feature of **hydrodynamic models** and we can try to check it experimentally (let's focus on **pions**).

EXPERIMENTAL UNIVERSAL SPECTRA FOR PIONS

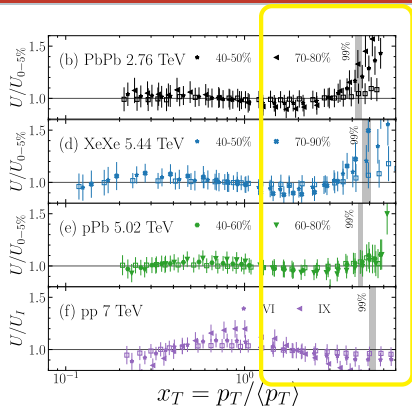
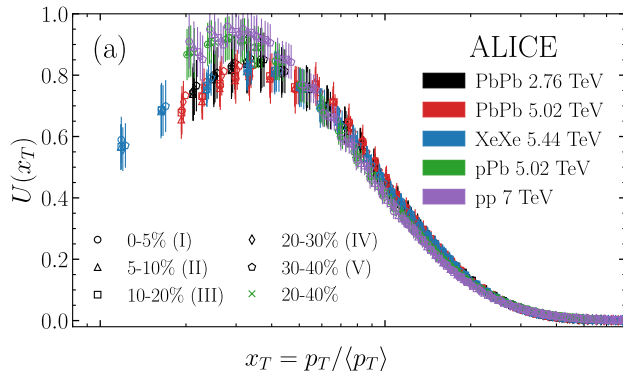


EXPERIMENTAL UNIVERSAL SPECTRA FOR PIONS



- **Universality over collision systems and centralities, for low p_T !**

EXPERIMENTAL UNIVERSAL SPECTRA FOR PIONS



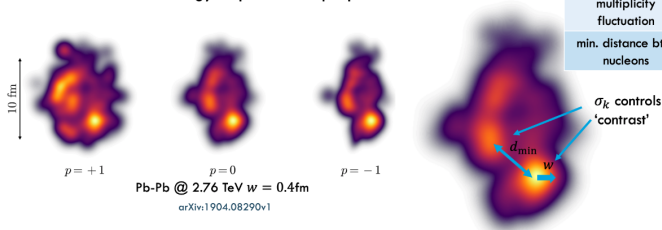
- **Universality over collision systems and centralities, for low p_T !**
- **Breaking of this universality at large p_T (onset of semi-hard physics)**

Let's focus on Pb-Pb 2.76 TeV

Model parameters
physical properties
from each stage
17 parameters

Initial Energy Deposition (TRENTO)

Parametrization for energy deposition at proper time $\tau = 0^+$

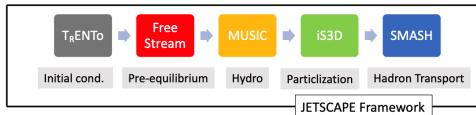
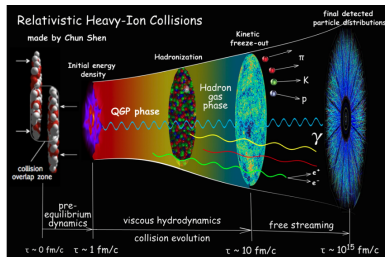


Parameter	Symbol
reduced thickness	p
nucleon width	w
energy normalization	N
multiplicity fluctuation	σ_k
min. distance btw. nucleons	$d_{\min}$

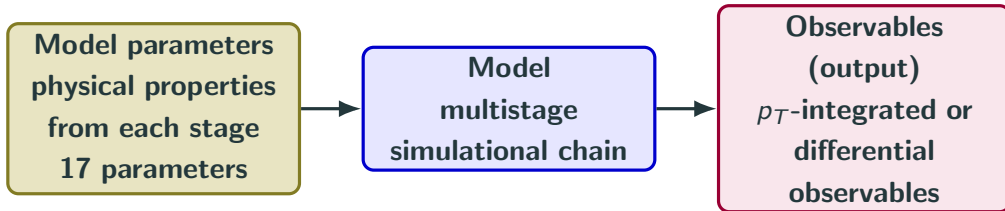
STATE-OF-THE-ART MODEL

Model parameters
physical properties
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17 parameters

Model
multistage
simulational chain



STATE-OF-THE-ART MODEL

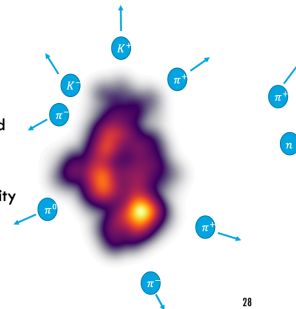


Observables

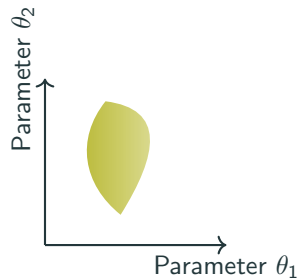
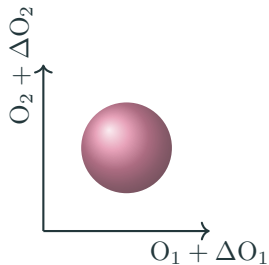
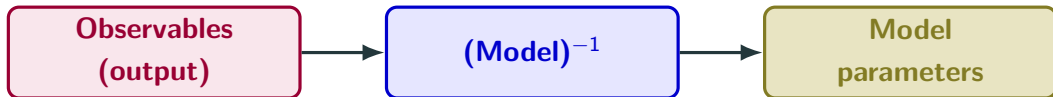
LHC Pb-Pb 2.76 TeV	RHIC Au-Au 200 GeV
dN_i/dy	dN_i/dy
$\langle p_T \rangle_i$	$\langle p_T \rangle_i$
$dN_{ch}/d\eta$	
$v_n\{2\}$	$v_n\{2\}$
$dE_T/d\eta$	
$\frac{\delta p_T}{\langle p_T \rangle}$	
$i \in \{\pi, K, p\}$	$i \in \{\pi, K\}$
$n \in \{2, 3, 4\}$	$n \in \{2, 3\}$

p_T -integrated
observables

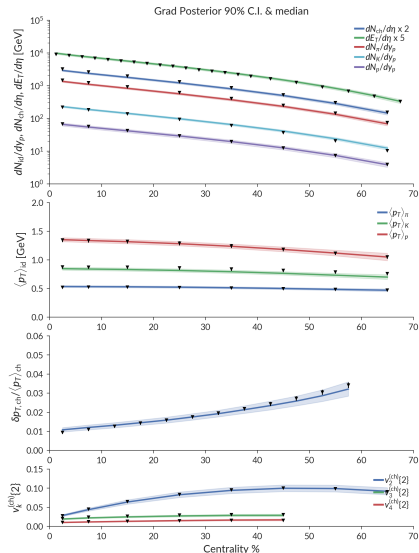
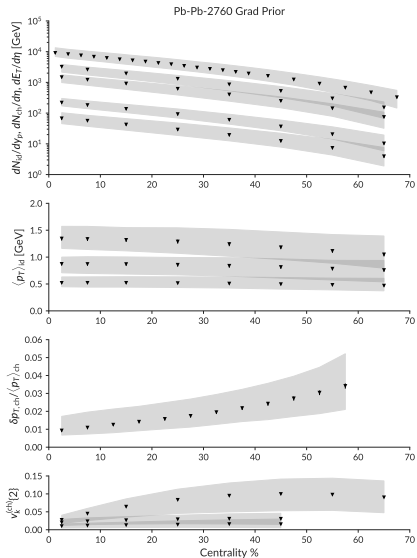
Same centrality
bins as
experiments



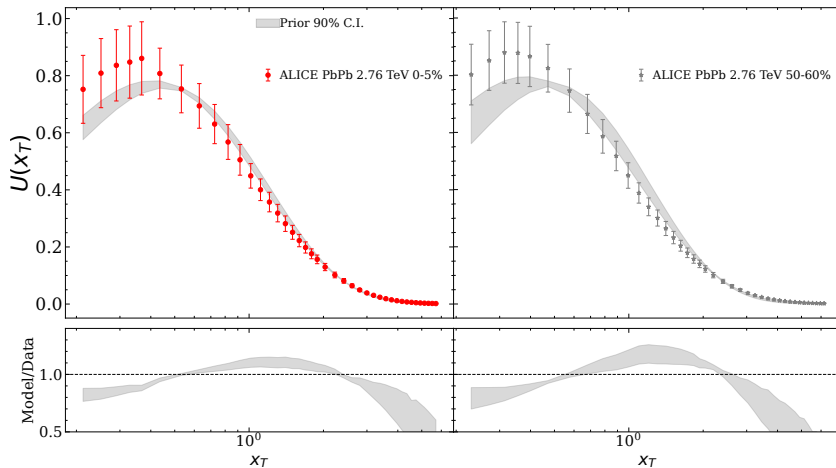
BAYESIAN ANALYSIS: THE INVERSE PROBLEM



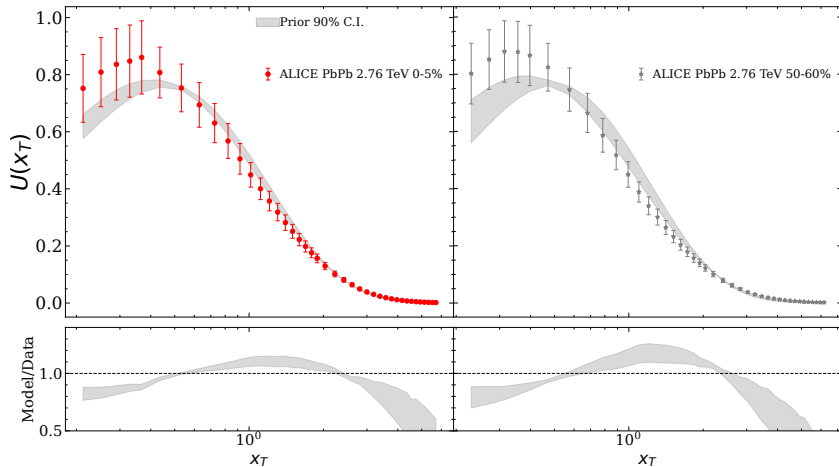
BAYESIAN ANALYSIS IN HEAVY-ION COLLISIONS: PRIOR + POSTERIOR



EXPLORING THE PARAMETER SPACE: SCALED SPECTRA PRIOR

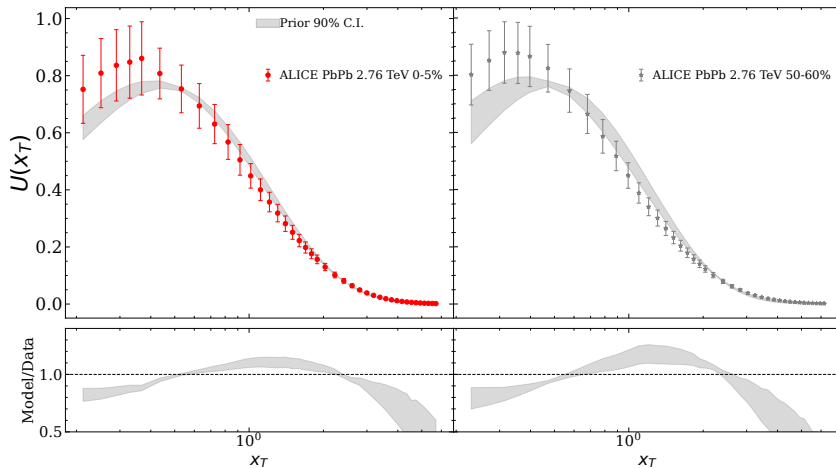


EXPLORING THE PARAMETER SPACE: SCALED SPECTRA PRIOR



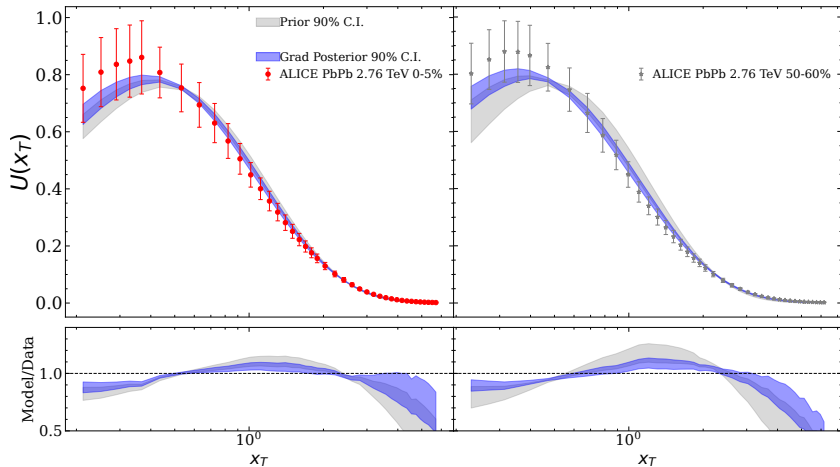
- The scaled spectra remains universal, even over a large variation in parameters;

EXPLORING THE PARAMETER SPACE: SCALED SPECTRA PRIOR



- The scaled spectra remains universal, even over a large variation in parameters;
- There is limited flexibility to tune the model to this observable!

UNIVERSAL SPECTRA CONSTRAINING POWER



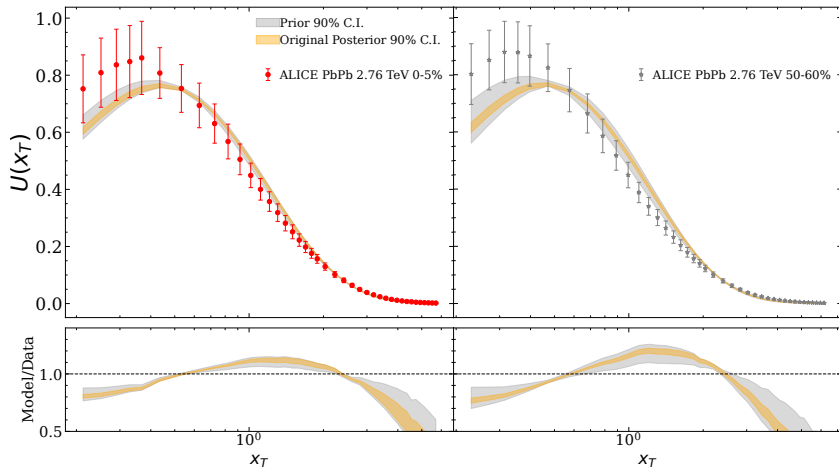
- **Scaled spectra calibration:** consistent with data within error bars

HOW ROBUST IS A BAYESIAN CALIBRATION?

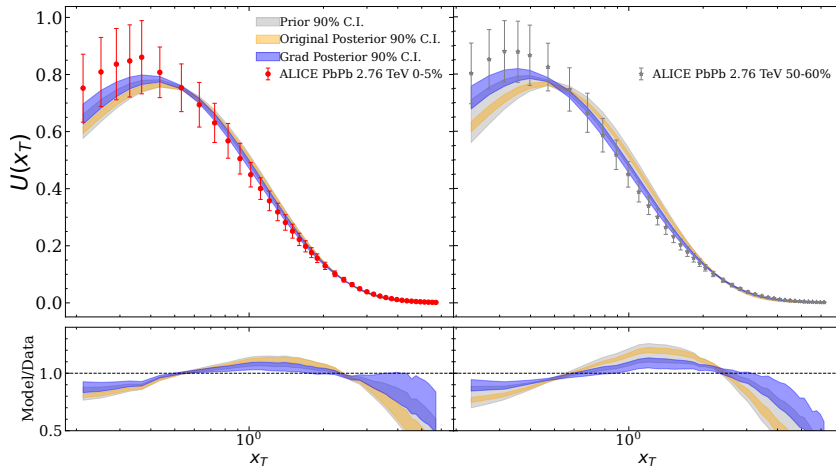
We can try to use the posterior from calibration to only integrated observables:

HOW ROBUST IS A BAYESIAN CALIBRATION?

We can try to use the **posterior from calibration to only integrated observables**:

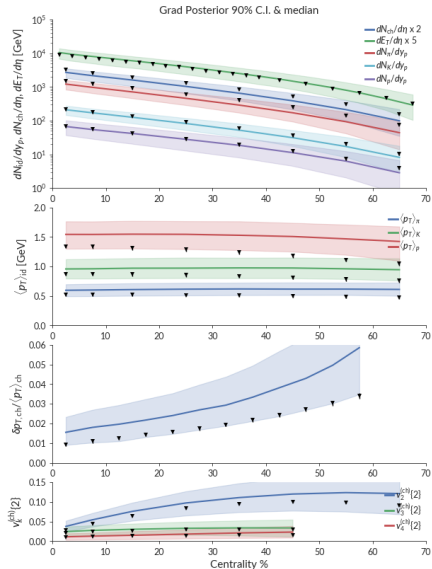


HOW ROBUST IS A BAYESIAN CALIBRATION?



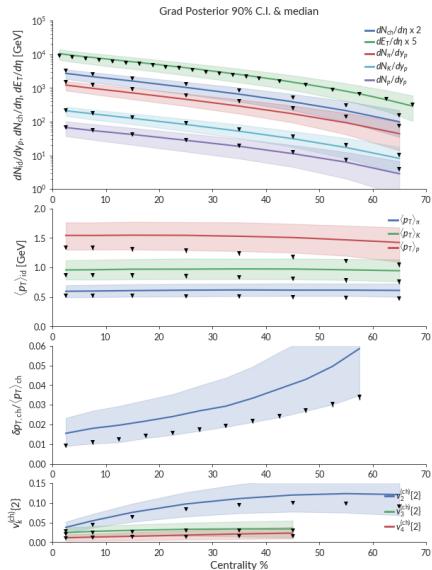
They are Inconsistent!

INTEGRATED OBSERVABLES WITH POSTERIOR FROM SPECTRA



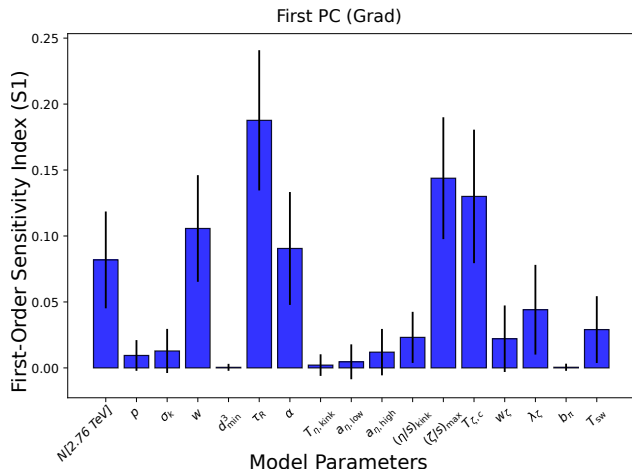
- Tension comes from $\langle p_T \rangle$

INTEGRATED OBSERVABLES WITH POSTERIOR FROM SPECTRA



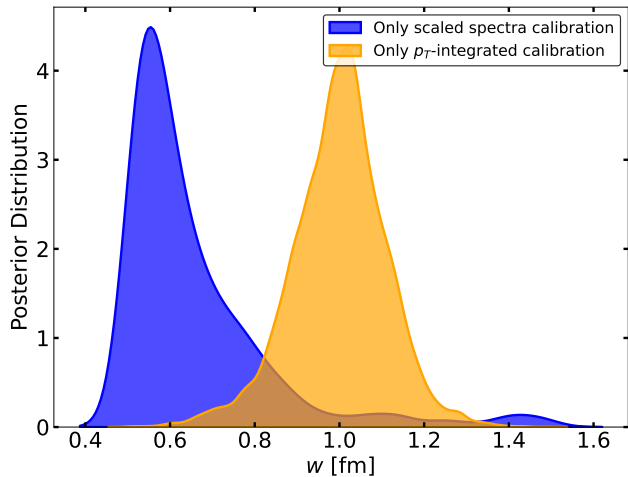
- Tension comes from $\langle p_T \rangle$
- What parameters cause this tension? → Sensitivity Global Analysis.

GLOBAL SENSITIVITY ANALYSIS

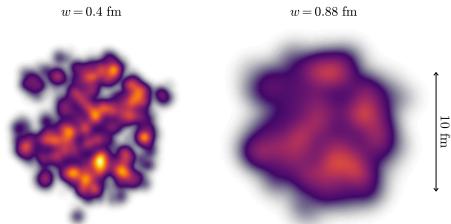
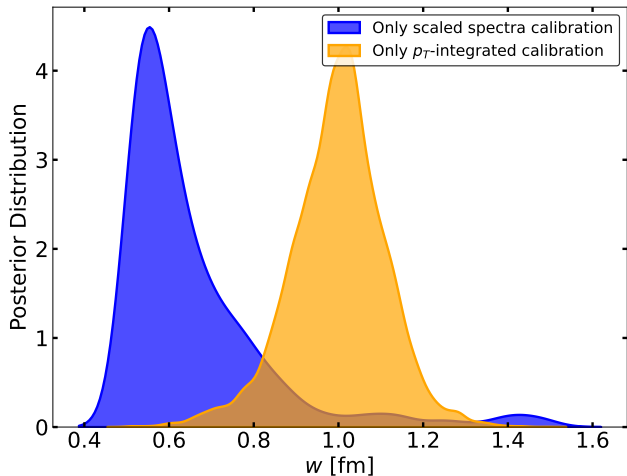


- Sensitive to bulk viscosity, time to initialize hydro, and w (nucleon width).

PARAMETER POSTERIORIORS: GAUSSIAN NUCLEON WIDTH

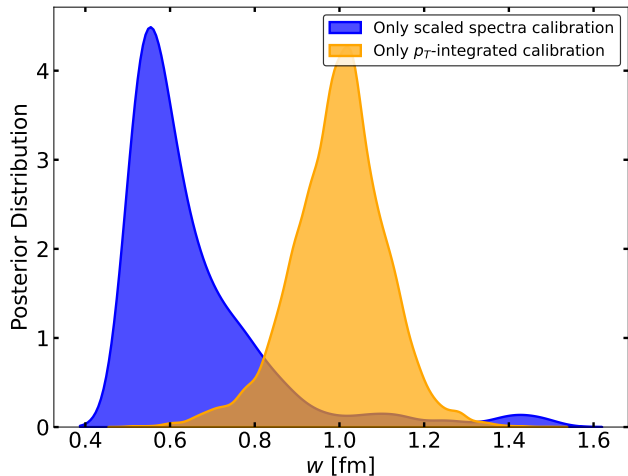


PARAMETER POSTERIORIORS: GAUSSIAN NUCLEON WIDTH



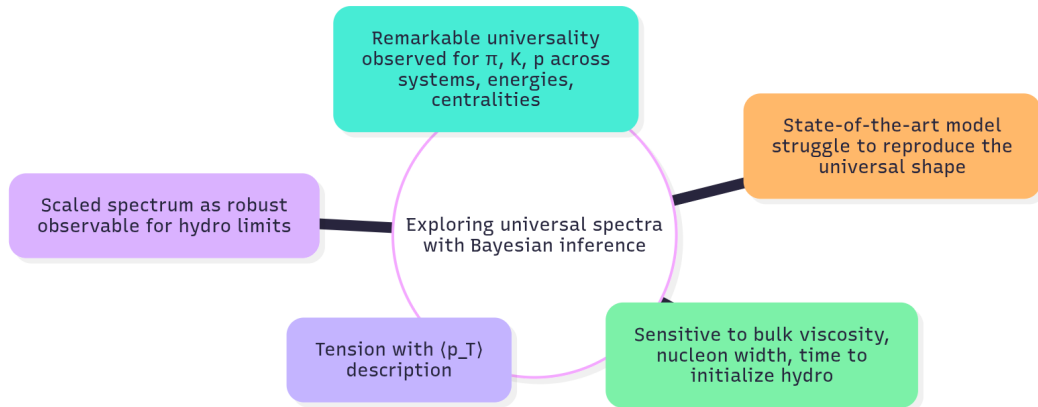
$$T_n(\mathbf{x}_\perp) = \frac{1}{2\pi w^2} \exp\left(-\frac{|\mathbf{x}_\perp|^2}{2w^2}\right)$$

PARAMETER POSTERIORIORS: GAUSSIAN NUCLEON WIDTH



- $U(x_T)$ needs small w
- $\langle p_T \rangle$ needs large w

MIND MAP: TAKE-HOME MESSAGES

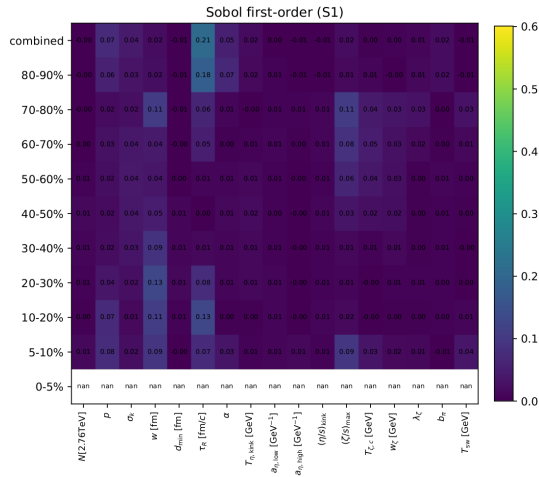


Resonance decays, collective flow, non-equilibrium contributions

FULL POSTERIOR DISTRIBUTION FOR ALL MODEL PARAMETERS

CORNER PLOTS FOR ALL MODEL PARAMETERS

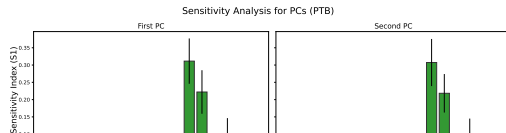
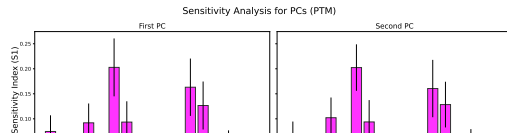
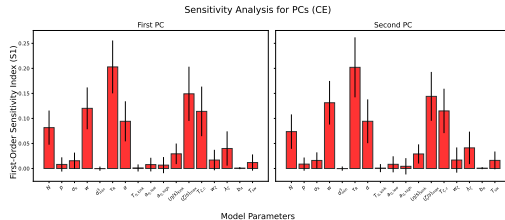
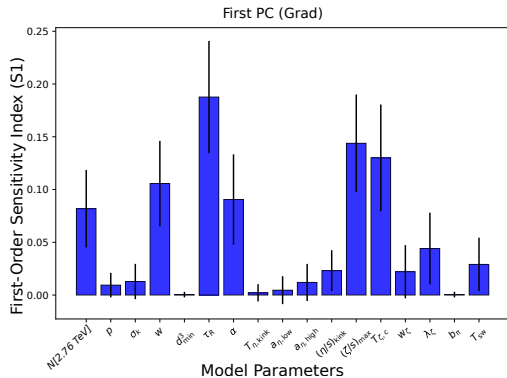
UNIVERSALITY OVER THE PARAMETER SPACE



- Bulk viscosity doesn't break universality
- The nucleon width (w) is the strongest contribution.

POSTERIOR PREDICTIONS AGAINST EXPERIMENTAL DATA: ONLY THE
UNIVERSAL SPECTRA CALIBRATION VS. PT-INTEGRATED OBSERVABLES
CALIBRATION

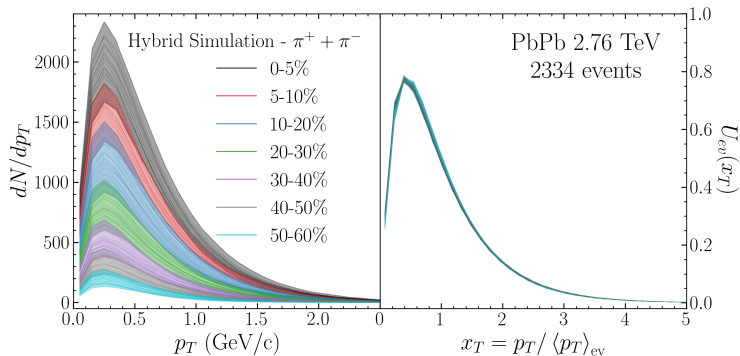
SENSITIVITY OVER DIFFERENT VISCOUS CORRECTIONS MODELS



UNIVERSAL SCALED SPECTRA: EVENT-BY-EVENT ANALYSIS

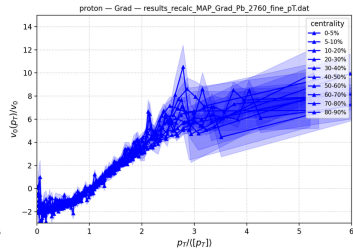
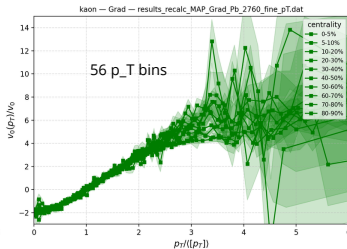
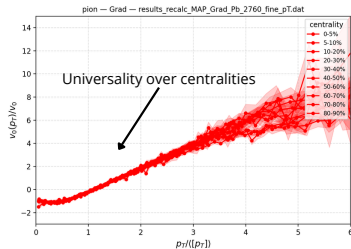
$$U_{\text{ev}}(x_T) \equiv \frac{\langle p_T \rangle_{\text{ev}}}{N_{\text{ev}}} \frac{dN_{\text{ev}}}{dp_T} = \frac{1}{N_{\text{ev}}} \frac{dN_{\text{ev}}}{dx_T}, \quad (2)$$

Event-by-event p_T spectra for Pb-Pb collisions at 2.76 TeV across different centralities



Universal behavior for each event, not just between centralities!

UNIVERSAL SCALED SPECTRA AND THE RADIAL DIFFERENTIAL FLOW

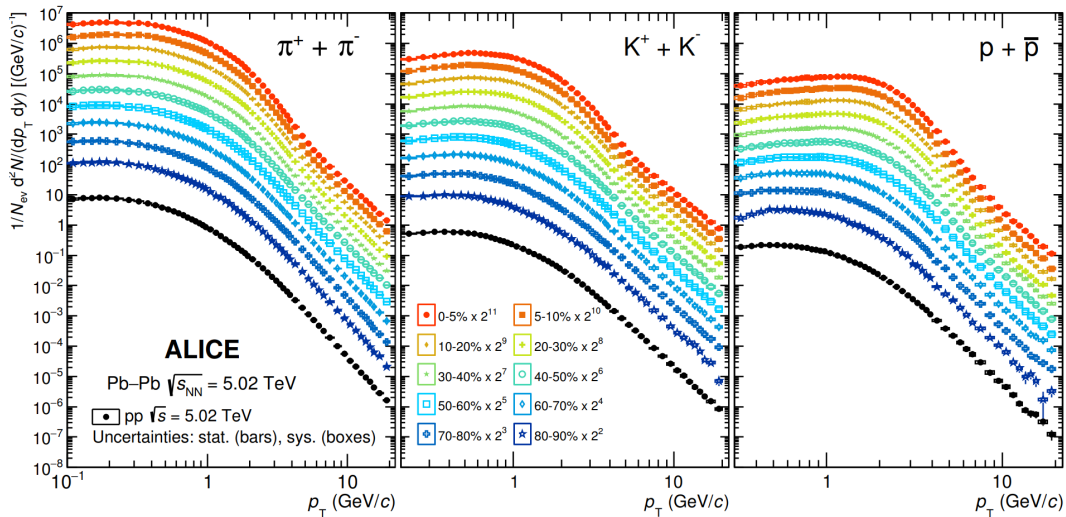


$$v_0(p_T) = \frac{\langle \delta n(p_T) \cdot \delta[p_T] \rangle}{\langle n(p_T) \rangle \cdot \sigma_{[p_T]}}$$

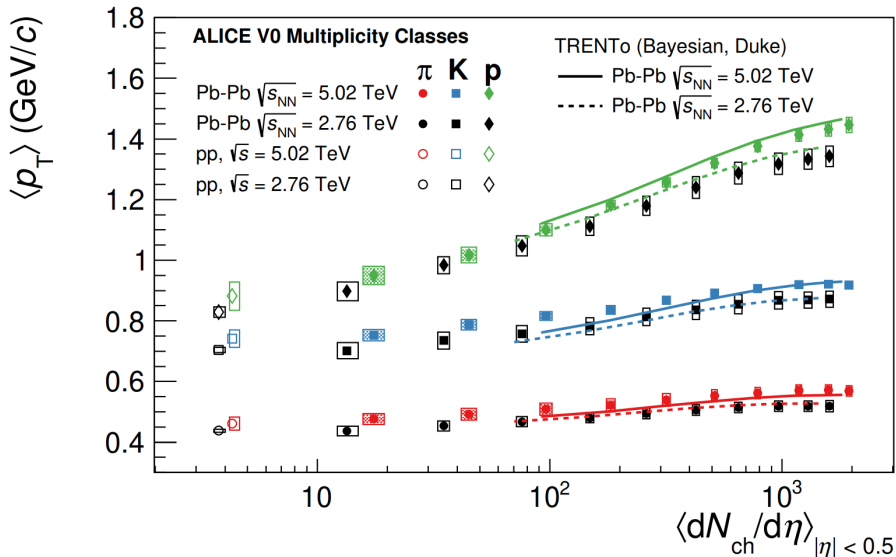
$$v_0 = \sigma_{[p_T]} / \langle [p_T] \rangle$$

$[p_T]$
event-wise mean- p_T

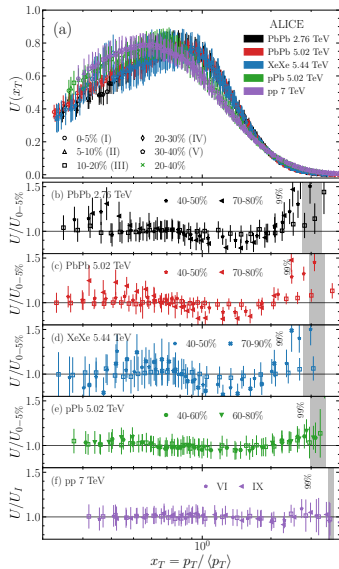
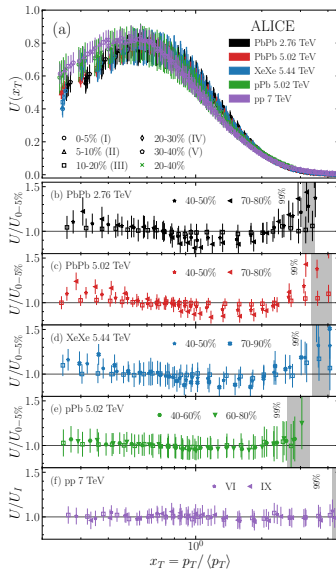
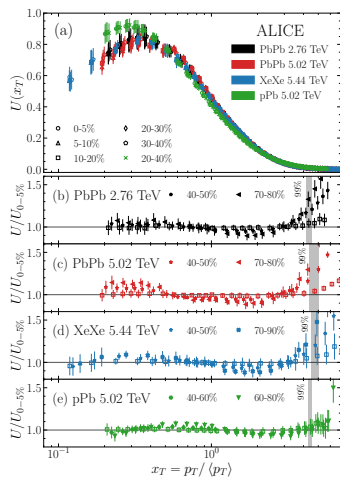
EXPERIMENTAL DATA: HOW TO OBTAIN THE UNIVERSAL SPECTRA



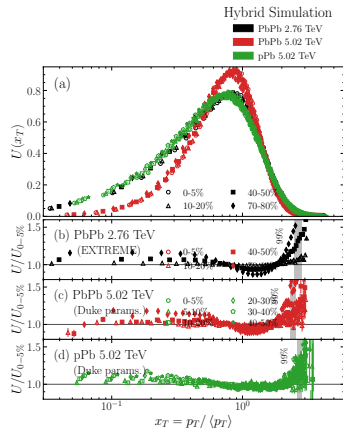
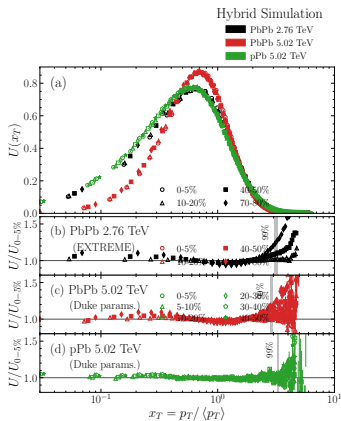
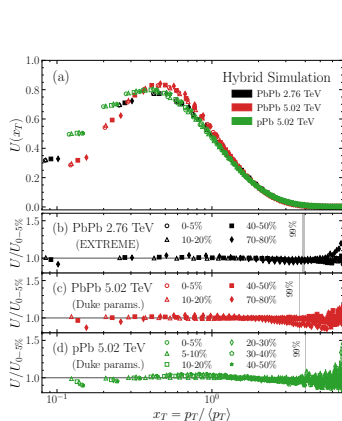
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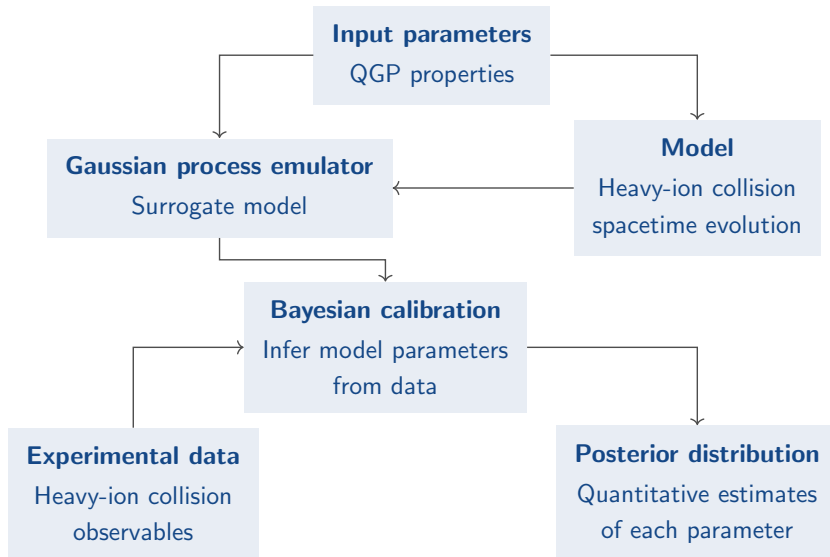
PIONS, KAONS, PROTONS - EXPERIMENT



PIONS, KAONS, PROTONS - THEORY



BAYESIAN ANALYSIS FLOWCHART: BPE PROCESS



JETSCAPE FRAMEWORK FLOWCHART

BAYESIAN INFERENCE SCHEMA

